Electro-Optic Modulators: Working Principles and Modern Advancements

NANENG 231: Electromagnetics
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***Abstract*—Every time you stream a video, ask an AI a question — your data travels as light. The devices making this possible are Electro-Optic Modulators (EOMs), which translate electrical signals into optical pulses at frequencies pushing past 100 GHz. In this project, we break down exactly how they work. Starting with Maxwell's equations, we move through the Pockels effect and into the mechanics of the Mach-Zehnder Interferometer. We then use that foundation to unpack a recent 2026 advancement in Thin-Film Lithium Niobate modulators. The main hurdle engineers face here is synchronizing a microwave signal with a light beam on the exact same chip to unlock next-generation bandwidth. By linking core physics to practical photonic engineering, this work takes a closer look at the physical infrastructure being built right now for 6G.**

I. Introduction

In modern electromagnetics applications and optical communications, it becomes vital to be able to convert high-frequency electrical signals to optical ones. An EOM becomes the main component in this process, since it enables converting high-speed electrical pulses into modulated optical signals. An EOM functions by varying the phase, frequency, amplitude or polarization of the light beam travelling through an optical material due to an applied electrical voltage.

In the present project, an emphasis will be made on Mach–Zehnder Interferometer (MZI) type EOM based on the linear electro-optic effect, also called the Pockels effect. The reason for choosing this particular device is its practical nature, as it represents a good model of what is learned in NANENG 231. The function of an EOM creates a connection between the behavior of dielectric materials under the influence of electric field, the propagation of an electromagnetic wave and Maxwell's equations. The refractive index of a specialized crystal such as lithium niobate can be changed due to an applied voltage. This change results in the phase shift of the propagating wave, which in turn is translated into a change of optical intensity by the MZI.

The rest of the report will cover both the theoretical background behind these phenomena as well as the modern application of these devices in engineering. In Section II the operational principle of MZI EOM will be described along with the relevant governing laws, operational principle and device parameters. In Section III a thorough mathematical analysis of the physics behind this device will be presented, starting with basic principles and Maxwell's equations. In Section IV a recent research article on high-efficiency TFLN EOMs published in 2026 will be reviewed in light of their latest engineering advances, which includes improved buffering layers and special electrodes.

II. Working Functionality and Underlying Principles

To understand how an Electro-optic Modulator works, we have to look at how an external electric field interacts with the electromagnetic wave propagating through a crystal. The device we are focusing on relies on two main mechanisms: changing the phase of light using the Pockels effect, and then converting that phase change into an intensity change using a Mach-Zehnder Interferometer (MZI) structure.

A. The Governing Law: The Pockels Effect

The basic physics involved in the operation of an EOM is defined by the linear electro-optic effect, also known as the Pockels effect. If a DC or low-frequency electric field (E) is applied on such non-centrosymmetric crystals, the arrangement of charges inside the crystal gets disturbed. Due to such disturbances, there is a change in the permittivity tensor of the crystal, which leads to a change in its refractive index (n).

For any wave propagating in such crystals, there is a proportionality between the change in the refractive index (Δn) of the crystal and the electric field applied. Such relationship can be given by the equation:

$$\Delta n = -\frac{1}{2} r n^3 E \tag{1}$$

where:

- n is the original, unperturbed refractive index of the crystal.
- r is the Pockels (electro-optic) coefficient, which depends on the specific material and the direction of the applied field relative to the crystal axes.
- E is the applied electric field, which can be approximated as $E = V/d$, where V is the applied voltage and d is the distance between the electrodes.

B. Phase Modulation

As noted in previous research on the propagation of waves, the phase of the electromagnetic wave depends on the refractive index of the material through which the wave is propagated. The wave experiences a phase change ($\Delta\phi$), due to the applied voltage, after propagation through a path of length L in the crystal. The phase change can be calculated as follows:

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta n L = -\frac{\pi r n^3 V L}{\lambda d} \tag{2}$$

where λ is the free-space wavelength of the optical wave. This equation shows that we can directly control the phase of the light by simply turning a voltage knob.

C. MZI Structure: Phase to Intensity Conversion

Despite the benefits provided by PM, in practice, almost all communication receivers are engineered to sense changes in the intensity of light as opposed to only phase. This is where the MZI comes in handy.

In an MZI, the incident optical wave is split into two paths having the same length. Then, an electric field is introduced into one path (or both paths in a push-pull fashion). During propagation within the paths, the applied voltage creates a phase shift between the two waves. Once the waves merge, interference occurs.

Depending on the created phase shift between the two interfering waves, the output interference may be either fully constructive (maximum output intensity) or fully destructive (zero output intensity). The intensity of the output wave, I_{out}, is calculated using the interference equation:

$$I_{out} = I_{in}\cos^2\left(\frac{\Delta\phi}{2}\right) \tag{3}$$

This beautifully simple relation allows the MZI to translate our electrical voltage signal directly into an optical on/off signal.

D. Key Parameter: The Half-Wave Voltage

When designing EOMs, the most important performance metric is the half-wave voltage, denoted as V_π. This is the exact voltage required to induce a phase shift of exactly π radians (180°). A phase shift of π means the waves from the two MZI arms will completely cancel each other out (destructive interference), switching the modulator from fully ”on” to fully ”off”.

By setting $\Delta\phi = \pi$ in Equation 2 and taking the magnitude, we get the equation for the half-wave voltage:

$$V_\pi = \frac{\lambda d}{r n^3 L} \tag{4}$$

From a design perspective, we want V_π to be as small as possible so that the device consumes less power. Equation 4 tells us that to lower the voltage, we need to either increase the interaction length (L), decrease the gap between electrodes (d), or find a material with a higher Pockels coefficient (r).

E. Physical Layout of the MZI Modulator

Before considering the derivation mathematically, it is beneficial to consider the actual physical structure of the device. Figure 1 shows the typical design of the Mach-Zehnder Interferometer electro-optic modulator. The functioning of the system can be broken down into four distinct physical components:

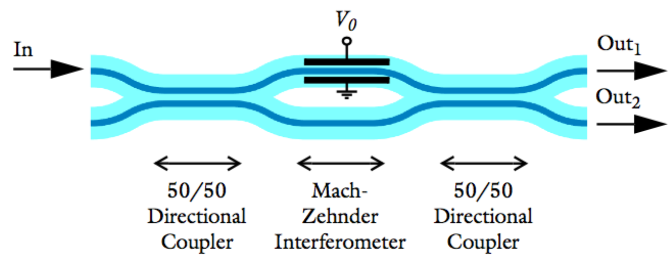

Fig. 1. Schematic representation of a Mach-Zehnder Interferometer (MZI) electro-optic modulator.

- **Optical Input:** The process starts on the left side of the apparatus where a CW laser beam is injected into the input waveguide. During this step, the light maintains constant amplitude and phase.
- **The Y-Splitter:** The input waveguide branches out into two paths. This Y-branch acts as an optical splitter, splitting the light equally into two beams with half the optical power traveling in each direction.
- **The Modulation Region (RF Electrodes):** This part experiences the electro-optic effect. RF metal electrodes run parallel to the optical waveguides. When an RF voltage is supplied to the electrodes, an electric field is produced within the crystal. In a standard ”push-pull” configuration, the electric field is directed in a way that the refractive index in one arm rises, while it falls in the other, producing a phase difference in the light beams in both arms.s.
- **The Combiner and Output (Energy Conservation):** After the propagation along the active region, both waveguides join together, thereby causing interference between the two light waves. Constructive interference takes place when the two wavefronts are exactly in phase, which causes maximum optical power to emanate from the main output (”on” state/logic ”1”). Meanwhile, a phase difference of 180° (π radians) between the two wavefronts caused by an input voltage results in destructive interference taking place from the primary output (”off” state/logic ”0”).
  It is crucial to note that all the processes mentioned above can take place against the background of the principle of energy conservation in electromagnetism, meaning that no electromagnetic energy can either be created or destroyed. Destructive interference in the main output channel (“0”) calls for the energy redistribution to maintain the energy conservation principle according to Poynting’s theorem. As for the directional coupler,

which has two outputs, then, according to the principle of energy conservation, maximum output power should be attained in the other channel and minimum output power in the first one and vice versa. In a basic single mode Y-combiner, on the contrary, no energy is absorbed or destroyed but dissipated into unguided modes.

By rapidly changing the voltage applied to the electrodes, the device continuously transitions between constructive and destructive interference, successfully mapping the high-speed electrical data onto the optical carrier wave.

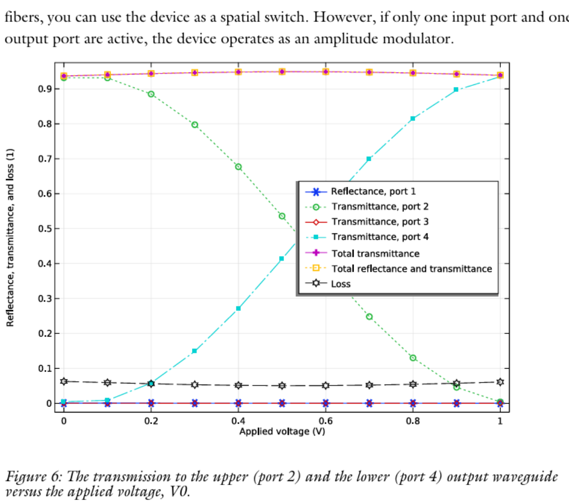

Fig. 2. 3D structural view of an MZI modulator with a directional coupler output. The presence of two distinct output ports (Output Fiber 1 and Output Fiber 2) allows the device to strictly adhere to energy conservation during destructive interference.

According to Fig. 2, modern implementations of the Mach-Zehnder interferometer (MZI) make use of a directional coupler at the output instead of a simple Y-splitter. If the RF input signal causes a phase change such that there is destructive interference, meaning no light is transmitted through Output Fiber 1, then equivalently, the interferometer will transmit that very amount of optical energy to Output Fiber 2. This example shows how Poynting's theorem is realized in practice.

III. Rigorous Mathematical Derivation

To rigorously understand the electro-optic modulation mechanism, we must start from first principles: Maxwell's equations in a dielectric medium. We assume a non-magnetic ($\mu = \mu_0$), source-free region ($\rho_{\text{free}} = 0$, $\mathbf{J} = 0$).

A. Maxwell's Equations and the Wave Equation

The fundamental Maxwell's equations governing the propagation of the optical field begin with Gauss's law for the electric field. In its most general form, the divergence of the total electric field $\mathbf{E}$ is proportional to the total volume charge density ρ_T:

$$\nabla \cdot \mathbf{E} = \frac{\rho_T}{\epsilon_0} \tag{5}$$

In a dielectric material, the total volume charge density is the sum of the free volume charge density (ρ_v) and the polarization volume charge density (ρ_{pv}), which is also known as the bound charge:

$$\rho_T = \rho_v + \rho_{pv} \tag{6}$$

The polarization volume charge density ρ_{pv} arises from the microscopic displacement of bound charges under an applied field, creating a macroscopic Polarization vector $\mathbf{P}$. The relationship is given by:

$$\rho_{pv} = -\nabla \cdot \mathbf{P} \tag{7}$$

Substituting this back into Gauss's law yields:

$$\nabla \cdot \mathbf{E} = \frac{\rho_v - \nabla \cdot \mathbf{P}}{\epsilon_0} \tag{8}$$

Rearranging the equation to group the vector fields together gives:

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_v \tag{9}$$

To simplify the macroscopic equations, we define the electric displacement field $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. This allows us to write Gauss's law strictly in terms of the free volume charge density:

$$\nabla \cdot \mathbf{D} = \rho_v \tag{10}$$

Assuming a source-free dielectric region for the optical wave propagation ($\rho_v = 0$, $\mathbf{J} = 0$) and a non-magnetic medium ($\mu = \mu_0$), the time-dependent Maxwell's equations are:

$$\nabla \cdot \mathbf{D} = 0 \tag{11}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{12}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \tag{13}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} \tag{14}$$

To derive the wave equation, we take the curl of Faraday's law (13):

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \tag{15}$$

Substituting Ampere's law (14) into the right side gives:

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \frac{\partial^2 \mathbf{D}}{\partial t^2} \tag{16}$$

We expand the left side using the standard vector identity:

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \tag{17}$$

For a linear, isotropic, and homogeneous dielectric, $\mathbf{D} = \epsilon \mathbf{E}$. Since $\nabla \cdot \mathbf{D} = 0$, it strictly follows that $\nabla \cdot \mathbf{E} = 0$. Equation (16) simplifies to the general wave equation:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{18}$$

To understand the speed of light and the refractive index, we first consider a vacuum where the permittivity is ϵ_0. The wave equation becomes:

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{19}$$

Comparing this to the standard 3D wave equation form ($\nabla^2 \mathbf{E} = \frac{1}{v^2}\frac{\partial^2 \mathbf{E}}{\partial t^2}$), we can rigorously prove that the speed of the electromagnetic wave in a vacuum, denoted as c, is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \tag{20}$$

Now, returning to our dielectric medium where $\epsilon = \epsilon_0 \epsilon_r$, the velocity v of the wave in this specific medium is:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon}} = \frac{1}{\sqrt{\mu_0 \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\epsilon_r}} \tag{21}$$

The optical refractive index n is fundamentally defined as the ratio of the speed of light in a vacuum to the speed of light in the medium ($n = c/v$). Substituting our velocity expression gives:

$$n = \frac{c}{c/\sqrt{\epsilon_r}} = \sqrt{\epsilon_r} \tag{22}$$

Squaring both sides rigorously proves the fundamental relationship between the optical refractive index and the relative permittivity of the material:

$$\epsilon_r = n^2 \tag{23}$$

Therefore, substituting this back, the final optical wave equation governing light propagation in the modulator is written as:

$$\nabla^2 \mathbf{E} = \frac{n^2}{c^2}\frac{\partial^2 \mathbf{E}}{\partial t^2} \tag{24}$$

B. Nonlinear Polarization and the Pockels Effect

In strongly driven non-centrosymmetric crystals like $LiNbO_3$, the linear approximation fails. We must expand the polarization vector $\mathbf{P}$ as a Taylor series in terms of the applied field:

$$\mathbf{P} = \epsilon_0 \left(\chi^{(1)} \mathbf{E} + \chi^{(2)} \mathbf{E}^2 + \chi^{(3)} \mathbf{E}^3 + \dots \right) \tag{25}$$

The Pockels effect arises from the second-order nonlinear susceptibility tensor $\chi^{(2)}$. To isolate this effect, we define the total electric field as the superposition of a static/low-frequency modulating field $\mathbf{E}_{DC}$ and a high-frequency optical carrier wave $\mathbf{E}_{\text{opt}}(t) = \mathbf{E}_0 \cos(\omega t)$:

$$\mathbf{E}(t) = \mathbf{E}_{DC} + \mathbf{E}_0 \cos(\omega t) \tag{26}$$

Substituting this total field into the second-order polarization term $\mathbf{P}^{(2)} = \epsilon_0 \chi^{(2)} \mathbf{E}^2$:

$$\mathbf{P}^{(2)} = \epsilon_0 \chi^{(2)} \left(\mathbf{E}_{DC}^2 + 2\mathbf{E}_{DC}\mathbf{E}_0 \cos(\omega t) + \mathbf{E}_0^2 \cos^2(\omega t) \right) \tag{27}$$

Applying the trigonometric identity $\cos^2(\omega t) = \frac{1}{2}(1 + \cos(2\omega t))$ expands the final term. The optical field propagating at frequency ω is strictly governed by the polarization components oscillating at ω. Isolating these terms from the full expansion $\mathbf{P} = \mathbf{P}^{(1)} + \mathbf{P}^{(2)}$ yields:

$$\mathbf{P}_\omega = \epsilon_0 \chi^{(1)} \mathbf{E}_0 \cos(\omega t) + 2\epsilon_0 \chi^{(2)} \mathbf{E}_{DC} \mathbf{E}_0 \cos(\omega t) \tag{28}$$

Factoring out the optical field yields an effective linear susceptibility:

$$\mathbf{P}_\omega = \epsilon_0 \underbrace{\left[\chi^{(1)} + 2\chi^{(2)} \mathbf{E}_{DC} \right]}_{\chi_{\text{eff}}} \mathbf{E}_{\text{opt}}(t) \tag{29}$$

C. Index Perturbation and Boundary Conditions

To map the nonlinear polarization back to macroscopic optical parameters, we return to the displacement field relation: $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$. This fundamental constitutive relation is universally valid in macroscopic electromagnetism assuming the medium has a strictly local response, meaning it exhibits no *spatial dispersion*. It is crucial to distinguish this from *temporal (chromatic) dispersion*. While $LiNbO_3$ is highly dispersive temporally (its refractive index heavily depends on the optical frequency ω), its spatial dispersion at optical frequencies is negligible. This allows us to strictly evaluate the polarization at any given point in space based solely on the electric field at that exact point.

Using the effective polarization derived from the Taylor expansion, we substitute $\mathbf{P}_\omega = \epsilon_0 \chi_{\text{eff}} \mathbf{E}_{\text{opt}}$. It may initially appear contradictory to use a linear-form equation when dealing with a strictly nonlinear material like $LiNbO_3$. However, this formulation is strictly valid under the **weak optical field constraint** ($|\mathbf{E}_{\text{opt}}| \ll |\mathbf{E}_{DC}|$). Because the optical field is infinitesimally weak compared to the applied modulating field, second-harmonic optical generation ($\mathbf{E}_{\text{opt}}^2$) is physically negligible. Therefore, the medium behaves as an *effectively linear* material from the perspective of the optical wave, where its effective susceptibility is artificially biased by the strong DC field.

Under this constraint, the displacement field for the optical wave becomes:

$$\mathbf{D}_\omega = \epsilon_0 \mathbf{E}_{\text{opt}} + \epsilon_0 \chi_{\text{eff}} \mathbf{E}_{\text{opt}} = \epsilon_0 (1 + \chi_{\text{eff}}) \mathbf{E}_{\text{opt}} \tag{30}$$

By the macroscopic definition of a linear dielectric, $\mathbf{D} = \epsilon_0 \epsilon_{r,\text{eff}} \mathbf{E}_{\text{opt}}$. Comparing these two expressions rigorously proves that the effective relative permittivity is:

$$\epsilon_{r,\text{eff}} = 1 + \chi_{\text{eff}} \tag{31}$$

From our previous proof in Subsection A, we established that the square of the refractive index is equal to the relative permittivity ($\epsilon_r = n^2$), strictly assuming a non-magnetic medium ($\mu_r = 1$). Therefore, the effective refractive index of the medium under the applied DC field is:

$$n_{\text{eff}}^2 = 1 + \chi_{\text{eff}} = 1 + \chi^{(1)} + 2\chi^{(2)} \mathbf{E}_{DC} \tag{32}$$

Knowing that the unperturbed index is $n^2 = 1 + \chi^{(1)}$, we can isolate the perturbation:

$$n_{\text{eff}}^2 = n^2 + 2\chi^{(2)} \mathbf{E}_{DC} \tag{33}$$

Let us define the perturbed refractive index as $n_{\text{eff}} = n + \Delta n$. To solve this, we must apply the **weak perturbation constraint** ($\Delta n \ll n$). Under this condition, we can truncate the binomial expansion of the square to the first order: $(n + \Delta n)^2 \approx n^2 + 2n\Delta n$. Equating the two expressions gives:

$$2n\Delta n = 2\chi^{(2)} \mathbf{E}_{DC} \implies \Delta n = \frac{\chi^{(2)}}{n} \mathbf{E}_{DC} \tag{34}$$

To align with standard optical engineering notation, we define the macroscopic Pockels electro-optic coefficient as $r =$

$-2\chi^{(2)}/n^4$. Substituting this definition yields the exact linear index change:

$$\Delta n = -\frac{1}{2} n^3 r \mathbf{E}_{DC} \tag{35}$$

From Equation 35, the modification of bulk properties can be mathematically modeled; nevertheless, the behavior of the equation of motion (EOM) is wholly determined by the electromagnetic boundary conditions. The optical signal travels through a waveguide structure consisting of a core region made of high refractive index materials ($n_1 = n_{\text{eff}}$), which is surrounded by a low-refractive-index cladding (n_2).

As we approach the dielectric interface separating the core and the cladding, there exist certain boundary conditions that result from Maxwell's Equations. There are two assumptions in this case: (i) the interface is a perfect dielectric without any free surface charge ($\rho_s = 0$) and (ii) there are no surface currents ($\mathbf{K} = 0$).

The basic boundary conditions below $\hat{n}$ imply the following definitions: **unit normal vector** is a unit normal to the interface pointing from the medium 2 (cladding) to the medium 1 (core). In mathematical manipulations on $\hat{n}$, the meaning is as follows: ($\hat{n}\times$) selects the component tangential to the interface, whereas ($\hat{n}\cdot$) selects the component normal to the interface.

Given these constraints, the fields must fulfill the requirement of:

Under these conditions, the fields must satisfy:

$$\hat{n} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0 \quad \text{(Tangential } \mathbf{E} \text{ is continuous)} \tag{36}$$
$$\hat{n} \times (\mathbf{H}_1 - \mathbf{H}_2) = 0 \quad \text{(Tangential } \mathbf{H} \text{ is continuous)} \tag{37}$$
$$\hat{n} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = 0 \quad \text{(Normal } \mathbf{D} \text{ is continuous)} \tag{38}$$
$$\hat{n} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0 \quad \text{(Normal } \mathbf{B} \text{ is continuous)} \tag{39}$$

If the wave is required to be strictly confined and guided inside the core by using the concept of total internal reflection, then the solutions to the wave equation should fulfill all the four boundary conditions simultaneously. Such mathematical condition limits the waves to travel in specific transverse modes only. Each mode has its unique longitudinal propagation constant, β.

By decomposing the electric fields, $\mathbf{E}_{\text{air}}$ and $\mathbf{E}_{\text{LN}}$, into orthogonal components tangential (t) and normal (n) to the boundary surface, the geometric projections yield:

$$E_{\text{air},t} = E_{\text{air}} \sin\theta_{\text{air}}, \qquad E_{\text{air},n} = E_{\text{air}} \cos\theta_{\text{air}}$$
$$E_{\text{LN},t} = E_{\text{LN}} \sin\theta_{\text{LN}}, \qquad E_{\text{LN},n} = E_{\text{LN}} \cos\theta_{\text{LN}}$$

Enforcing the continuity of the tangential electric field across the interface, $\hat{n} \times (\mathbf{E}_{\text{air}} - \mathbf{E}_{\text{LN}}) = 0$, requires that:

$$E_{\text{air}} \sin\theta_{\text{air}} = E_{\text{LN}} \sin\theta_{\text{LN}} \tag{40}$$

Concurrently, the absence of free surface charge dictates that the normal component of the electric flux density, $\mathbf{D} = \epsilon\mathbf{E}$, must remain continuous ($\hat{n} \cdot (\mathbf{D}_{\text{air}} - \mathbf{D}_{\text{LN}}) = 0$). This constraint produces:

$$\epsilon_{\text{air}} E_{\text{air}} \cos\theta_{\text{air}} = \epsilon_{\text{LN}} E_{\text{LN}} \cos\theta_{\text{LN}} \tag{41}$$

Taking the ratio of Equation 40 to Equation 41 effectively eliminates the field magnitudes, establishing a direct relation between the relative permittivities and the deflection angles:

$$\frac{E_{\text{air}} \sin\theta_{\text{air}}}{\epsilon_{\text{air}} E_{\text{air}} \cos\theta_{\text{air}}} = \frac{E_{\text{LN}} \sin\theta_{\text{LN}}}{\epsilon_{\text{LN}} E_{\text{LN}} \cos\theta_{\text{LN}}}$$

Simplifying this expression through standard trigonometric identities yields the fundamental refraction law for the electric field at the boundary:

$$\tan\theta_{\text{air}} = \left(\frac{\epsilon_{\text{air}}}{\epsilon_{\text{LN}}}\right) \tan\theta_{\text{LN}} \tag{42}$$

An analogous mathematical treatment applies to the magnetic field, $\mathbf{H}$, provided there are no surface currents present. If ϕ_{air} and ϕ_{LN} denote the respective angles of the magnetic field vectors relative to the interface normal, applying the continuity of tangential $\mathbf{H}$ and normal $\mathbf{B}$ ($\mathbf{B} = \mu\mathbf{H}$) leads to a symmetric set of relations:

$$H_{\text{air}} \sin\phi_{\text{air}} = H_{\text{LN}} \sin\phi_{\text{LN}} \tag{43}$$
$$\mu_{\text{air}} H_{\text{air}} \cos\phi_{\text{air}} = \mu_{\text{LN}} H_{\text{LN}} \cos\phi_{\text{LN}} \tag{44}$$

Dividing Equation 43 by Equation 44 mirrors the prior derivation, isolating the angular dependencies in terms of magnetic permeability:

$$\tan\phi_{\text{air}} = \left(\frac{\mu_{\text{air}}}{\mu_{\text{LN}}}\right) \tan\phi_{\text{LN}} \tag{45}$$

For an electromagnetic wave to be strictly confined and guided within the core (total internal reflection), the solutions to the wave equation must satisfy these four boundary conditions simultaneously. This mathematical constraint ensures that the wave can only propagate in discrete transverse modes. Each guided mode is characterized by a specific longitudinal propagation constant, β.

The propagation constant controls the variation of the spatial phase along the direction of propagation, which is z-direction, according to the exponential phasor, $e^{-j\beta z}$. In the fundamental mode approximation, the propagation constant is defined using the core's refractive index and the free space wavenumber ($k_0 = 2\pi/\lambda$):

$$\beta = n_{\text{eff}} k_0 = n_{\text{eff}} \frac{2\pi}{\lambda} \tag{46}$$

As the wave propagates over the physical length of the modulator electrodes (L), it accumulates a total phase of $\phi = \beta L$. When the applied voltage changes the core index by Δn, the boundary conditions shift, which in turn alters the propagation constant by $\Delta\beta$:

$$\Delta\beta = \Delta n k_0 = \Delta n \frac{2\pi}{\lambda} \tag{47}$$

The total induced phase shift at the output of the modulator is simply the change in the propagation constant multiplied by the interaction length:

$$\Delta\phi = \Delta\beta L = \frac{2\pi}{\lambda} \Delta n L \tag{48}$$

Finally, substituting our rigorously derived expression for Δn (Equation 35) directly into this phase equation yields the fundamental transfer function of the phase modulator:

$$\Delta\phi = \frac{2\pi}{\lambda}\left(-\frac{1}{2}n^3 r\mathbf{E}_{DC}\right)L = -\frac{\pi n^3 r\mathbf{E}_{DC}L}{\lambda} \tag{49}$$

This rigorous proof bridges the gap between microscopic nonlinear polarization, macroscopic Maxwell boundary conditions, and the final engineered phase shift of the device.

D. Energy Conservation: Poynting's Theorem

Finally, to rigorously address the power distribution at the output directional coupler (referenced in Section II), we employ Poynting's Theorem. The differential form states:

$$-\nabla\cdot\mathbf{S} = \frac{\partial u}{\partial t} + \mathbf{J}\cdot\mathbf{E} \tag{50}$$

where $\mathbf{S} = \mathbf{E}\times\mathbf{H}$ is the Poynting vector, and $u = \frac{1}{2}(\mathbf{E}\cdot\mathbf{D}+\mathbf{B}\cdot\mathbf{H})$ is the electromagnetic energy density. Integrating over the volume V of the output coupler and applying the Divergence Theorem:

$$-\oint_{\partial V}\mathbf{S}\cdot d\mathbf{A} = \int_V \frac{\partial u}{\partial t}dV + \int_V \mathbf{J}\cdot\mathbf{E}dV \tag{51}$$

In a lossless dielectric ($\mathbf{J} = 0$) under steady-state continuous-wave operation ($\langle\partial u/\partial t\rangle = 0$), the net time-averaged power flux through the closed surface must be zero:

$$\oint_{\partial V}\langle\mathbf{S}\rangle\cdot d\mathbf{A} = 0 \tag{52}$$

This mathematical condition ensures that the optical power input P_{in} is equal to the sum of the optical power outputs $P_{\text{out},1}$ and $P_{\text{out},2}$. The interference being destructive in port 1 guarantees constructive interference in port 2 in order to maintain the zero sum of fluxes, thus directly connecting Maxwell's macroscopic energy conservation law to the functionality of the device.

IV. Modern EOM Design

In our earlier sections, we spent a lot of time on the fundamental math behind Electro-Optic Modulators (EOMs). But how do engineers actually build these things for real-world, high-bandwidth networks like 6G? To bridge the gap between abstract theory and physical reality, we took a look at a recent 2026 paper by Xue *et al.*, titled *"High-performance thin-film lithium niobate electro-optic modulator on a thick silica buffering layer."*

A. The Core Issue: Managing Two Waves at Once

Modern modulators don't just use a flat DC voltage. To encode data, they use a rapidly oscillating microwave (RF) signal. This means the device has to simultaneously handle two completely different electromagnetic waves traveling down the exact same physical path:

- **The Optical Carrier:** A laser beam confined inside the dielectric Lithium Niobate (LN) waveguide.
- **The Modulating Microwave:** An RF signal traveling along the metal electrodes.

As you can see in the flowchart (Figure 3), getting these two waves to interact without ruining the signal is a massive boundary-value problem. Xue *et al.* successfully tackled the physical limitations, but as we'll discuss at the end, there's still room to optimize the device using pure electromagnetic theory.

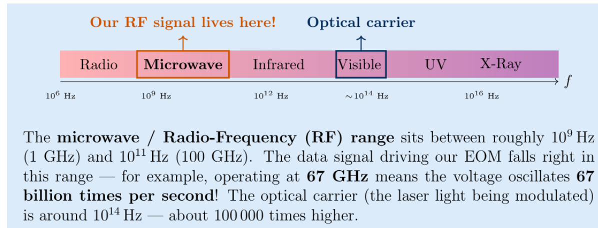

Fig. 3. A flowchart outlining the electromagnetic hurdles in EOM design, the physical fixes used by Xue *et al.*, and our proposed boundary-condition fix to maximize field overlap.

B. The Walk-Off Effect and Structural Fixes

For the microwave to properly modulate the light via the Pockels effect, it has to constantly squeeze the optical waveguide. But because of chromatic dispersion, the microwave's phase velocity (v_m) and the optical wave's group velocity (v_g) are naturally different.

If $v_m \neq v_g$, the two waves eventually drift apart as they travel down the chip. The alternating positive and negative cycles of the RF wave start to destructively interfere with the optical phase shift, effectively canceling out the signal. This is known as the **Walk-Off effect**. To fix it, engineers have to force $v_m = v_g$.

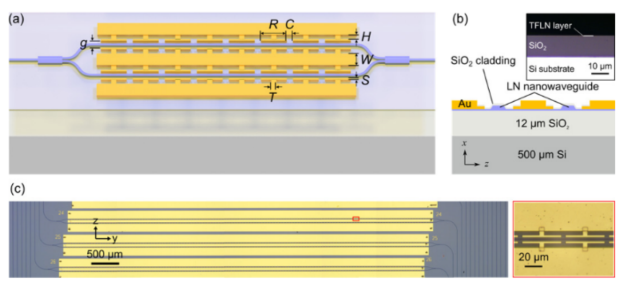

Fig. 4. (a) Schematic of the EOM. (b) Cross-section showing the thick silica buffer beneath the waveguide. (c) Top-down microscope image of the periodic T-rails.

Achieving this velocity match is tough on standard silicon chips because silicon has a high relative permittivity ($\epsilon_r \approx 11.9$), which severely slows down the microwave. Looking at Figure 4, Xue *et al.* solved this using two structural changes:

1. The Silica Buffer Layer: They inserted a 12-μm-thick layer of silica ($\epsilon_r \approx 3.9$) between the LN waveguide and the silicon base (Figure 4b). This lower-permittivity barrier speeds the microwave back up, pushing its velocity closer to the optical wave.

2. T-Rail Electrodes: To get a strong electric field ($E = V/d$), you need a tiny gap between the electrodes. But a narrow gap causes high-frequency RF currents to bunch up at the edges (the skin effect), burning up power. The researchers used Capacitively Loaded Traveling-Wave (CLTW) electrodes—basically adding periodic "T-rails" into the gap (Figure 4c). Because these teeth are spaced out, the current stays in the wider main lanes, cutting resistance by 30%. Plus, the teeth act like tiny shunt capacitors. By tweaking their size, the engineers could mathematically dial in the distributed capacitance to perfectly match the microwave velocity to the optical velocity.

V. Simulation

To evaluate the performance of the electro-optic modulator, the device geometry was first modeled using finite-element and finite-difference eigenmode solvers. The core structure consists of a thin-film lithium niobate (TFLN) layer between a lower buffer and an upper cladding, with a ground-signal-ground (GSG) electrode configuration placed on top.

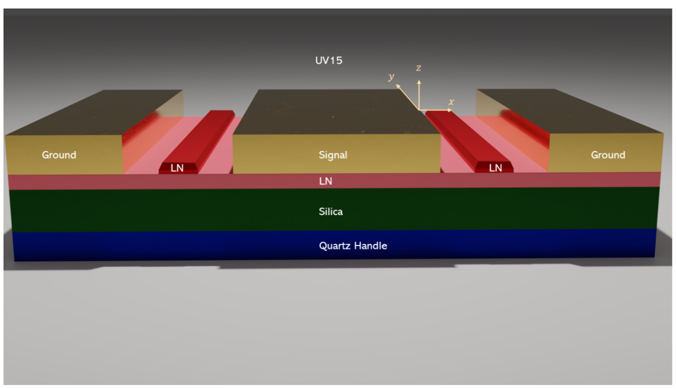

Fig. 5. Multi-planar layout view of the thin-film lithium niobate (TFLN) electro-optic modulator.

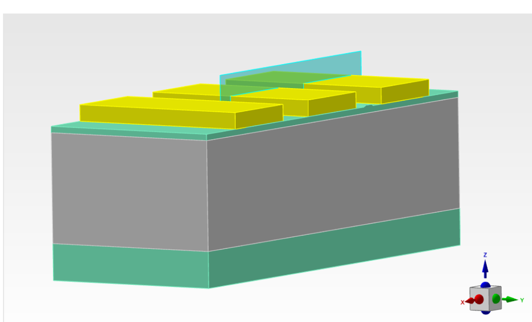

Fig. 6. Three-dimensional schematic of the thin-film lithium niobate (TFLN) electro-optic phase modulator geometry.

Figure 6 illustrates the three-dimensional schematic of the simulated device, highlighting the distinct material layers and the electrode layout.

A. Simulation results

To determine how the applied voltage changes the material properties of the modulator, an electrostatic simulation was performed. The solver calculates the static electric field (E) generated between the electrode plates with a voltage equals 5 V.

Figure 7 displays the spatial distribution of the electrostatic electric field within the lithium niobate core when a 5 V signal is applied. Because the electrode gap is on the micrometer scale, the resulting electric field intensity reaches the order of 10^6 V/m, with highly localized hotspots crowding at the metal edges.

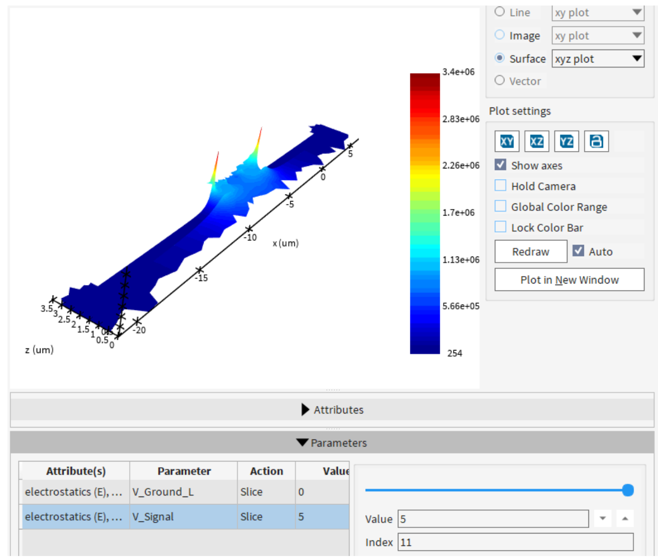

Fig. 7. Spatial distribution of the electrostatic electric field (V/m) inside the TFLN waveguide core induced by a 5 V signal.

To determine the actual electro-optic index perturbation (dn), this calculated electric field (E) is applied to the Pockels effect equation:

$$\Delta n = -\frac{1}{2} n^3 r_{33} E \tag{53}$$

Because the linear electro-optic coefficient (r_{33}) for lithium niobate is exceptionally small—typically in the range of 30×10^{-12} m/V—multiplying it by the large 10^6 V/m electric field results in a highly controlled, fractional change in the refractive index.

Figure 8 displays the resulting spatial change in the refractive index (dn) calculated from this electric field interaction.

As the voltage shifts, the mesh vertices map the localized variations in dn. This visualization confirms that the maximum index perturbation is on the order of 10^{-4} and is strictly confined to the active lithium niobate layer, providing the necessary phase-shifting mechanism required for the optical mode before it interacts with the high-frequency RF drive.

To evaluate the modulation efficiency of the 12 μm silica buffer design, the effective refractive index (n_{eff}) of the fundamental optical mode was extracted as a function of the applied driving voltage.

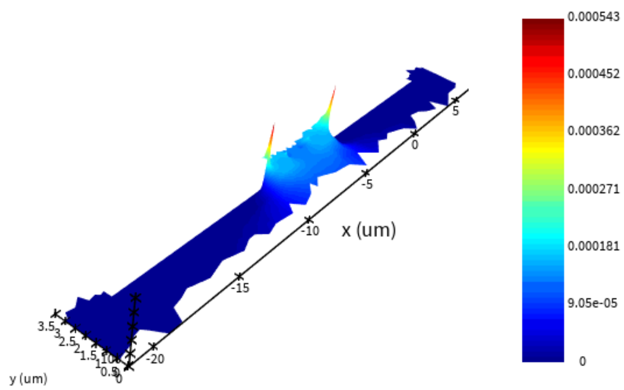

Fig. 8. Spatial distribution of the electro-optic index perturbation (dn) inside the TFLN waveguide core, peaking at approximately 5.4×10^{-4}.

Figure 9 plots this linear relationship. Crucially, the trendline does not pass through the origin; at 0 V, the y-intercept is approximately 2.02. This baseline value simply represents the natural, intrinsic effective index of the unperturbed TFLN waveguide.

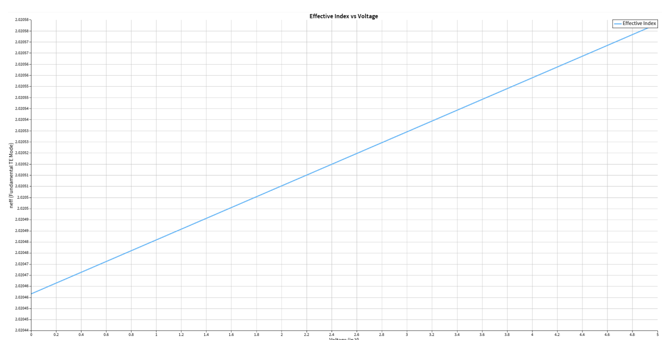

Fig. 9. Linear dependence of the effective index on the applied voltage due to the Pockels effect.

When voltage is applied, the effective index shifts linearly due to the linear electro-optic (Pockels) effect. The physics dictating this exact shift is defined by the following relationship:

$$\Delta n_{\text{eff}} = \frac{1}{2} n_e^3 r_{33} \frac{V}{G} \Gamma$$

Where n_e is the extraordinary refractive index, r_{33} is the dominant electro-optic tensor coefficient of lithium niobate, V is the applied voltage, G is the electrode gap, and Γ is the overlap integral between the optical and microwave fields.

To evaluate the practical modulation efficiency, the effective refractive index (n_{eff}) of the fundamental optical mode was extracted as a function of the applied driving voltage. Figure 8 plots this linear relationship, starting from the intrinsic zero-voltage baseline of approximately 2.02.

The curve shown in Figure 10 is governed by the following exact expression for L_π:

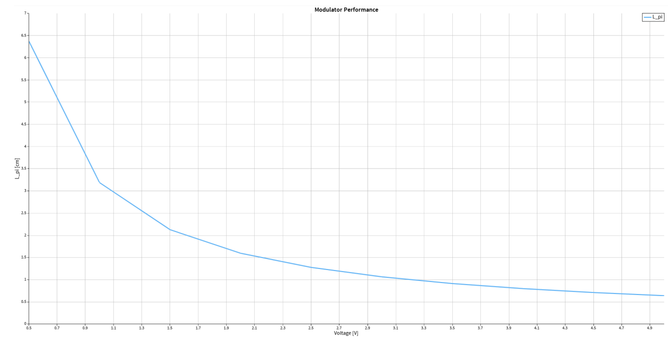

Fig. 10. Calculated π-phase-shift length (L_π) as a function of applied driving voltage, demonstrating the characteristic $1/V$ hyperbolic dependence.

$$L_\pi = \frac{\lambda}{2 \times \Re(n_{\text{eff}}(0\text{V}) - n_{\text{eff}}(V))}$$

To prove this relationship mathematically, we begin with the fundamental formula for the accumulated optical phase shift ($\Delta\phi$) over a given propagation length (L):

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta n_{\text{eff}} L$$

By definition, the half-wave length (L_π) is the specific propagation distance required to induce a phase shift of exactly $\Delta\phi = \pi$. Substituting π into the phase shift equation gives:

$$\pi = \frac{2\pi}{\lambda} \Delta n_{\text{eff}} L_\pi$$

Dividing both sides of the equation by π simplifies the expression to:

$$1 = \frac{2}{\lambda} \Delta n_{\text{eff}} L_\pi$$

The change in the effective index (Δn_{eff}) is defined as the difference in the real part of the index between the unperturbed state and the driven state, yielding $\Re(n_{\text{eff}}(0\text{V}) - n_{\text{eff}}(V))$. Substituting this definition into the simplified equation and rearranging to solve for L_π provides the final proof of the governing equation:

$$L_\pi = \frac{\lambda}{2 \times \Re(n_{\text{eff}}(0\text{V}) - n_{\text{eff}}(V))}$$

To fully characterize the modulator's driving efficiency, the half-wave voltage-length product ($V_\pi L$) was calculated by multiplying the required phase-shift length (L_π) by the applied voltage. Figure 11 plots this crucial performance metric.

Alongside the driving metrics, the propagation losses (in dB/cm) suffered by the fundamental TE mode were evaluated. This optical loss is extracted directly from the imaginary part of the effective index (κ) using the following standard expression:

$$\text{Loss} = -0.20 \times \log_{10}\left(e^{-2\pi\kappa/\lambda}\right)$$

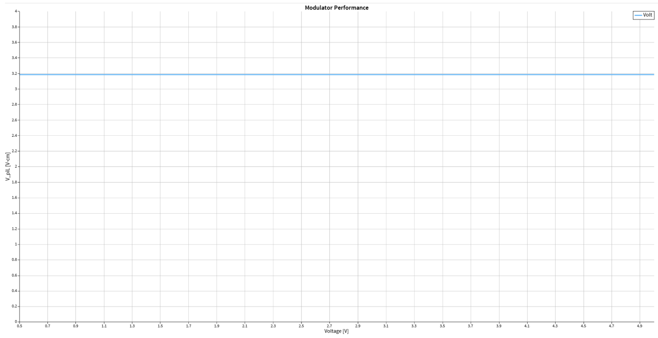

Fig. 11. Calculated half-wave voltage-length product ($V_\pi L$) as a function of applied voltage.

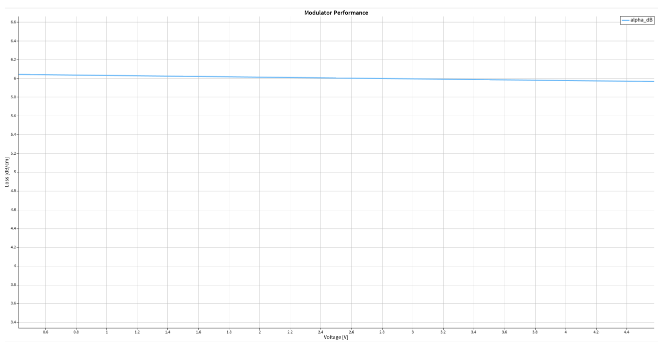

Fig. 12. Optical propagation loss of the fundamental mode versus applied voltage, demonstrating a constant loss profile.

Figure 12 displays the calculated optical loss across the driving voltage range.

The optical losses in the modulator remain almost perfectly constant regardless of the applied voltage. This indicates that the losses do not depend upon the Pockels effect; instead, they are primarily attributed to the evanescent tails of the optical mode interacting with the lossy gold (Au) electrodes. Consequently, the key geometric factor influencing this insertion loss is the separation gap between the electrodes, which must be carefully optimized to achieve the best ratio between high modulation performance and low optical absorption.

B. Silica Buffer Thickness Optimization

To justify the 12 μm silica buffer thickness used in this design, a MATLAB simulation was developed to study how changing the buffer thickness affects the speed of the microwave signal inside the electrode. The key requirement is simple: the microwave signal and the light pulse must travel at the same speed through the modulator. If they drift apart, the modulation quality degrades at high frequencies and the bandwidth suffers.

1) How the Simulation Works: The electrode sits on top of five stacked layers of different materials. Each material slows down the microwave signal by a different amount, depending on its permittivity. Silicon has a very high permittivity ($\varepsilon = 11.9$), so it slows the microwave down significantly. The SiO_2 buffer sits between the electrode and the silicon, shielding the

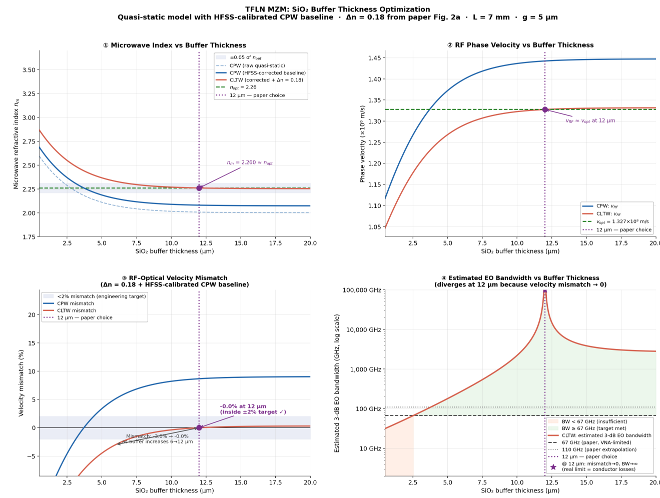

Fig. 13. MATLAB simulation showing how the SiO_2 buffer thickness affects microwave propagation in the TFLN modulator. The dashed vertical line marks the chosen 12 μm buffer thickness.

microwave field from the silicon. A thicker buffer means less silicon influence, which means a lower microwave refractive index n_m and a faster microwave signal.

The microwave index was calculated by splitting the total electric field energy into each layer and taking a weighted average of all the permittivities. The velocity mismatch was then defined as:

$$\delta = \frac{v_{\text{RF}} - v_{\text{opt}}}{v_{\text{opt}}} \times 100\% \tag{54}$$

where $v_{\text{RF}} = c_0/n_m$ is the microwave speed and $v_{\text{opt}} = c_0/2.26$ is the optical speed. The target is $\delta = 0\%$.

2) Results: Figure 13 shows four plots, each telling part of the same story:

- **Top-left — Microwave Index:** At thin buffers the silicon substrate dominates and the microwave index is high, reaching around 2.4–2.5 for the CLTW electrode near 0.5 μm. As the buffer thickens, the SiO_2 shields the field from the silicon and the index steadily drops. At 12 μm, the CLTW index reaches 2.26, matching the optical group index.
- **Top-right — RF Phase Velocity:** This is the direct consequence of the index plot. A lower index means a faster microwave. The CLTW curve meets the optical velocity target (dashed line) at 12 μm, confirming velocity matching.
- **Bottom-left — Velocity Mismatch:** This shows the percentage speed difference between the microwave and the light. The shaded band is the acceptable zone ($< 2\%$). The CLTW curve starts at large negative mismatch (RF too slow) and climbs to $\approx 0\%$ at 12 μm, landing inside the target band.
- **Bottom-right — Electro-Optic Bandwidth:** Bandwidth and velocity mismatch are inversely related — when the mismatch is large, bandwidth is low. As the mismatch

approaches zero near 12 μm, the theoretical bandwidth spikes sharply (log scale is used to show this). In the real device, conductor losses cap the bandwidth at a finite value, which is why the measured result is > 67 GHz rather than infinity.

In summary, 12 μm is the thickness where the CLTW electrode and the silica buffer together bring the microwave speed into alignment with the optical speed, maximizing the modulator bandwidth.

VI. Conclusion

This project started with Maxwell's equations and ended at a chip smaller than a fingernail pushing data at 67 GHz. The chain connecting them is surprisingly clean.

The Pockels effect gives us a voltage-controlled refractive index. That index shift, derived rigorously from the second-order susceptibility $\chi^{(2)}$ of $LiNbO_3$, is tiny — our simulation confirmed a peak perturbation of only 5.4×10^{-4} at 5 V. That local index perturbation, accumulated over the $L = 7\,\mathrm{mm}$ electrode length, is sufficient to produce a full π-radian phase shift at the operating voltage, switching the MZI from fully "on" to fully "off".

So the optical side of the problem is essentially solved by classical wave physics.

The harder problem is speed. Once you replace the DC voltage with a 67 GHz microwave signal, the RF wave and the light pulse have to travel at the exact same velocity across the entire chip, or the modulation washes out. Silicon's high permittivity ($\varepsilon_r \approx 11.9$) makes this very difficult by default because it slows the microwave down far below the optical group velocity. Xue *et al.* fixed this with two moves: a 12 μm SiO_2 buffer that shields the microwave field from the silicon substrate, and CLTW T-rail electrodes that fine-tune the remaining capacitive loading.

Our MATLAB simulation showed that this combination brings the CLTW velocity mismatch to approximately 0% at $12\,\mu$m — whereas the bare CPW electrode, lacking the slow-wave capacitive loading, undershoots the optical group index in the microwave index domain and overshoots into positive mismatch territory, confirming that both structural elements are jointly necessary. This velocity match at $12\,\mu$m is exactly where the measured bandwidth exceeds 67 GHz.

One limitation worth noting is that the optical insertion loss in our simulation stayed flat regardless of the applied voltage. This tells us the loss is controlled by electrode geometry, not by the electro-optic interaction itself. That is a boundary condition built into the platform, and it will remain a constraint for any traveling-wave modulator built this way.

Overall, what makes this device interesting is not any single innovation but the combination: good material choice ($LiNbO_3$), smart interferometer design (MZI), and careful electromagnetic engineering (buffer layer and CLTW electrodes) all working together.

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